

In Vitro Detection of Cavitation Induced by a Diagnostic Ultrasound System

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Abstract—Experiments were performed *in vitro* to investigate the possibility of transient cavitation from short-pulse, diagnostic ultrasound under conditions comparable to clinical ultrasound examinations. An acoustic backscattering technique that utilizes 30-MHz pulsed ultrasound was employed to probe the potential of cavitation production by a clinical ultrasound scanner, specifically a Hewlett-Packard 77020A. Two calibrated, phased-array HP imaging transducers with 2.5- and 5.0-MHz operating frequencies were driven in M-mode (single cycle) and Doppler mode (4 cycles) by the HP imaging system. Hydrophobic polystyrene spheres with an average diameter of $0.245\ \mu\text{m}$ and Albunex, an echo contrast agent consisting of 1- to $10\text{-}\mu\text{m}$ bubbles with a human serum albumen coating, served as potential cavitation nuclei. Cavitation was detected in the water with polystyrene spheres at 2.5 MHz in both M-mode and Doppler mode at a peak negative acoustic pressure of 1.1 MPa or greater. Insonification at 5.0 MHz in either mode did not produce a detectable amount of cavitation, even with peak negative pressures as high as 1.2 MPa. Also, cavitation was not detected in water with the Albunex spheres at either frequency.

I. INTRODUCTION

THE POSSIBILITY of cavitation, the rapid growth and violent collapse of a gas-filled cavity, as a consequence of short-pulse diagnostic ultrasound was predicted theoretically in 1982 by Apfel [1] and Flynn [2]. Acoustic cavitation inception is defined by a specific threshold: the minimum acoustic pressure necessary to initiate the growth of a microbubble, or nucleus, in a fluid during the rarefaction phase of the pressure cycle. Experimental measurements in various fluids have elucidated the importance of ultrasound, host fluid, and nuclei parameters in determining *in vitro* cavitation thresholds [3]–[6]. Child *et al.* measured threshold pressures for hemorrhage in mouse lung exposed to 1- to 4-MHz short-pulse ($<10\ \mu\text{s}$) diagnostic ultrasound [7]. The damage was linked to the presence of transient cavitation. Their data is the first to provide direct evidence that clinically relevant, pulsed ultrasound exposures produce deleterious effects in mammalian tissue in the absence of significant heating. However, epidemiologic

surveys and widespread clinical usage over 25 years have yielded no evidence of adverse human biological effects from diagnostic ultrasound. Ziskin and Petitti reason statistically that subtle effects, long-term delayed effects, and certain genetic effects could easily escape detection in studies with limited or biased sample populations [8].

In this study experiments were performed to determine whether a clinical diagnostic scanner, a Hewlett-Packard 77020A, could produce cavitation in water containing suspensions of either $0.245\text{-}\mu\text{m}$ polystyrene spheres or Albunex,[®] 1– $10\ \mu\text{m}$ albumin-coated microbubbles. These choices of working media, although clinically irrelevant, allow the host fluid and nuclei characteristics to be carefully controlled and monitored. The output of the HP scanner is within the guidelines set by the Federal Drug Administration and is typical of clinical diagnostic ultrasound equipment [9]–[11]. Transient cavitation was detected with an acoustic backscattering technique that utilizes $10\text{-}\mu\text{s}$ long, 30-MHz center frequency tone bursts that scatter from cavitation microbubbles [6]. This “active” cavitation detector (ACD), affords sufficient sensitivity and temporal resolution to detect noninvasively the presence of a single micron-sized bubble that lasts only a microsecond.

II. EXPERIMENTAL APPARATUS

The experimental apparatus is composed of a test chamber, a fluid management system, a Hewlett-Packard 77020A ultrasound imaging system, and an active cavitation detection system. The fluid management system, a closed, recirculating filtration network described elsewhere [5], delivers filtered and degassed water to the test chamber, or cavitation cell, which is diagrammed in Fig. 1. The test cell consists of a 3.4-l Plexiglass enclosure filled with the test liquid that is irradiated by two confocally positioned transducers. The clinical diagnostic ultrasound transducer, the Hewlett-Packard sector, phased-array transducer, was inserted into the tank via a rectangular opening in the cover and held rigidly in place using a freestanding clamp arrangement (not shown). A port in the side of the cell supported a narrow-band, 30-MHz, pulse-echo cavitation detection transducer. Acoustic tone bursts or single cycle pulses emanating from each transducer insonified overlapping volumes of fluid. Cavitation activity, or short-lived bubbles, induced by the HP transducer scattered some of the 30-MHz signal, resulting in a readily observable, backscattered signal. In order to detect these short-lived bubbles, they must

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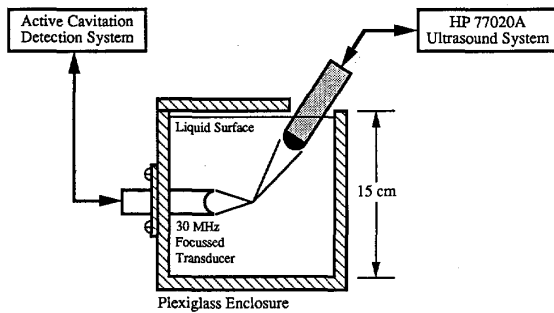


Fig. 1. Front view of test cell. The test liquid is filtered and slightly degassed water containing either 1) $0.245\ \mu\text{m}$ polystyrene spheres or 2) Alburnex.

have grown by at least a factor of two during the rarefaction phase and therefore in all likelihood could be considered "transient cavities" as in Flynn's original definition [12], [13]. This arrangement provided a means for real-time detection of transient cavitation activity.

The HP clinical scanner employed in this study was capable of operating in both the B-scan (i.e., imaging) and pulsed Doppler mode. Two transducers with operating frequencies of 2.5 and 5.0 MHz were utilized. The acoustic pulse amplitude and the pulse repetition rate could also be adjusted. The scanner pulse duration was fixed at approximately one and four cycles for the imaging and Doppler modes, respectively. In addition, panel controls allowed adjustment of the direction of the beam axis, a feature that facilitated the transducer alignment process. Although adjustable, the focal length of the scanner was kept fixed at 6.6 and 3.3 cm for the 2.5- and 5.0-MHz transducers, respectively. This choice of focal lengths ensured that the two transducer outputs were maximal. (See discussion under Section II-A.)

A series of experiments was performed to test whether or not the HP imaging system could produce short-lived cavitation *in vitro* at 2.5 and 5.0 MHz in either the M-mode (single cycle) or Doppler mode (four cycles). Two different particles were used as potential cavitation nuclei, $0.245\text{-}\mu\text{m}$ diameter polystyrene spheres and Alburnex, an echo contrast agent that consists of 1- to $10\text{-}\mu\text{m}$ air bubbles in a human serum albumin shell.

Cavitation detection was accomplished using the high-frequency active cavitation detector (ACD) diagrammed in Fig. 2. Roy *et al.* have given a detailed description of the design, implementation, and capabilities of this pulse-echo detector [6]. The ACD instrumentation produced a 30-MHz center frequency, $10\ \mu\text{s}$ long detector pulse that was synchronized with the scanner burst and delayed to ensure that the ACD and HP scanner pulses arrived in the confocal region simultaneously. This delay, which takes into consideration the focal length of the ACD transducer (2.5 cm) and of the HP transducer, was adjusted to $21\ \mu\text{s}$ for the 2.5-MHz HP transducer and $2\ \mu\text{s}$ for the 5.0-MHz HP transducer. Echoes returning from the ACD transmit/receive switch were fed into a LeCroy 9450 digital oscilloscope that created a record of the short-lived backscattering events produced by cavitation bubble activity.

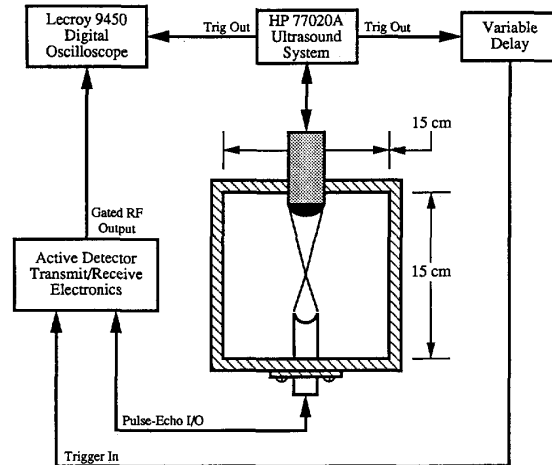


Fig. 2. Top view of test cell and block diagram of experimental apparatus.

Because transient cavitation events are short-lived, often lasting only as long as the tone bursts that cause the bubble growth, the digital oscilloscope was used in the sequential trigger mode in order to "capture" 20 traces that could be subsequently viewed at a later time. The 20 traces, or "bins," are not necessarily contiguous in time because the oscilloscope triggered only when the backscattered signal exceeded a preset trigger level (80 mV). This trigger level was chosen so that the oscilloscope triggered only when the backscattered signal exceeded the background within a specified time window, which corresponded to the focal region of the ACD. Provided the backscattered signal exceeded 80 mV, consecutive events occurring at the pulse repetition rate of the HP ultrasound system could be captured by the oscilloscope. Thus adjacent bins may or may not be aligned contiguously in time. For much of the data presented in this paper the digitizing rate was 40 Msamples/s. Because the ACD carrier frequency was 30 MHz, significant aliasing resulted. (As expected, the predominant frequency evident in the digital waveforms presented below is 10 MHz.) The envelope of the backscattered signal from these cavitation events can be interpreted roughly as bubble radial amplitude versus time curves [14]. Note, however, that sampling error could contribute to some of the observed structure. Cavitation bubble growth and collapse time is the same as the pulse duration of the HP ultrasound system. Thus, the gross features of the envelope of the scattered signal were fairly well preserved. This recording scheme, although limited, provided sufficient fidelity to support a simple detection criterion: the presence versus the absence of cavitation. This criterion was established through visual observation of the digitized record.

For all experiments the host water was open to the room air and seeded with polystyrene microspheres or Alburnex. Thus, one had to be prepared to distinguish cavitation bubble events from "background" echoes resulting from scattering by polystyrene microspheres or Alburnex microbubbles. On the time scale of the ACD pulse duration ($10\ \mu\text{s}$), particles moved slowly enough ($\sim 8\ \text{cm/s}$) to remain in the focal zone of the

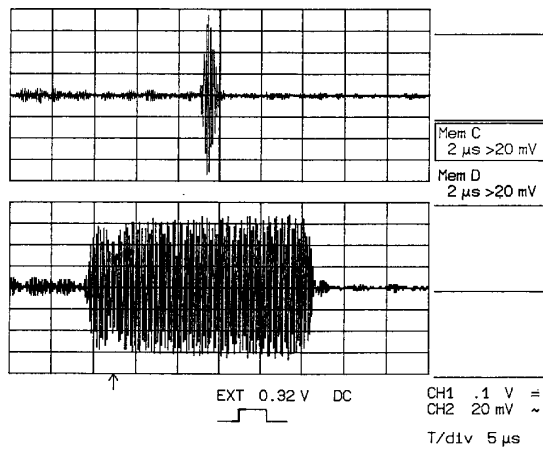


Fig. 3. Backscattered signals received by the active cavitation detector. The top trace indicates the presence of a short-lived cavitation event. The bottom trace indicates the presence of an aggregate of polystyrene spheres that persisted in the focal region of the detector for the entire pulse duration (10 μ s). The time base for the two traces is 2 μ s/div.

transducer during the entire ACD interrogation. On the other hand, cavitation events occurred over time scales that never exceeded 2 μ s. The best descriptor was not the echo amplitude alone, but the echo amplitude combined with its duration. Shown in Fig. 3 are two oscilloscope traces that illustrate the backscattered signals received by the active cavitation detector. The top trace is considered as the presence of a cavitation event and the bottom trace indicates the presence of what probably amounts to an aggregate of polystyrene spheres that persisted in the focal region of the detector for the entire detector pulse duration (10 μ s). Only two types of scatterers can result in such a signal as pictured on the top trace of Fig. 3: 1) a short-lived scatterer or 2) a scatterer moving at high velocity through the sensitive region of the detector. Since the -3 -dB beamwidth of the ACD transducer was of order 200 μ m, only particles moving at a velocity of ~ 100 m/s in a direction perpendicular to the detector beam axis could generate such an echo. Since neither convection currents, radiation force nor streaming mechanisms can produce such a speed [6], we conclude that the object that is scattering the 30-MHz tone burst only exists for the duration of the excitation pulse from the HP unit. For this reason, perturbations such as the one shown in the top trace of Fig. 3 were taken to be indicative of short-lived cavitation activity. In addition, the durations of the observed echoes were comparable to the pulse durations for the two scanner modes employed. The duration of the backscattered signal serves to discriminate between cavitation events that occur on the time scale of either a single acoustic cycle or the entire tone burst utilized by the HP scanner. Scattering from the polystyrene spheres produces a backscattered signal of the same duration as the 30-MHz tone burst (10 μ s). Backscattered signals of much shorter duration, 2 μ s or less, are attributed to individual cavitation bubbles.

Calibration

Absolute peak pressure calibration in the focal region of

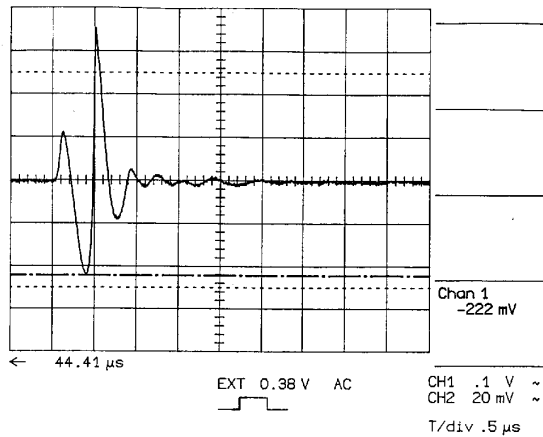
both HP transducers was achieved with a Marconi bilaminar polyvinylidene fluoride (PVDF) hydrophone that has a 1 mm active element. Figs. 4 and 5 contain oscilloscope traces of the received PVDF signals for the 2.5- and 5.0-MHz HP transducers, respectively. The pulse repetition frequency that yielded the highest peak negative pressure from the scanner was 2.83 kHz for the 2.5-MHz transducer and 5.0 kHz for the 5.0-MHz transducer. Note the nonlinear distortion caused by propagation through 6.6 and 3.3 cm of water, respectively. Evidence for ultrasonic finite-amplitude distortion in human tissue using medical equipment has been demonstrated by Starritt *et al.* [15]. More recently, Siddiqi *et al.* have measured distorted waveforms produced by transabdominal diagnostic transducers via a transvaginal hydrophone [16]. The peak negative acoustic pressure at the maximum power setting of the scanner was 1.3 MPa at 2.5 MHz and 1.2 MPa at 5.0 MHz in both M-mode and Doppler mode. The -3 -dB beam widths were ~ 7.5 mm for the 2.5-MHz transducer while focussed at a distance of 6.6 cm and ~ 1.6 mm for the 5.0-MHz transducer while focussed at a distance of 3.3 cm.

III. EXPERIMENTAL PROCEDURE

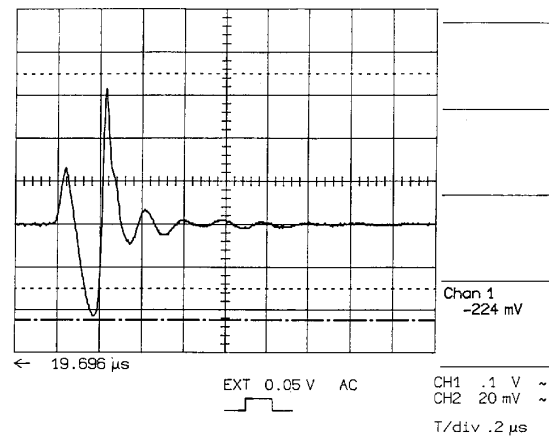
Alignment of the HP and ACD transducers is achieved using a target that consists of a 4-mm brass bead threaded on a thin wire. First the 30-MHz detection transducer is driven in pulse-echo mode (with a pulse duration of 10 μ s, corresponding to 300 cycles), and the received signal is maximized by positioning the target, which is mounted on a triple-axis micromanipulator rig. Then, without moving the target, the HP imaging system is driven in M-mode, the transducer position is adjusted, and the beam is steered so that the amplitude of the reflected signal is maximized on the M-mode trace. Care is taken to duplicate the focal distance utilized during calibration of the transducer. The target is then removed and the cavitation test cell is drained.

The test cell is then filled with partially degassed water that has been filtered in the closed loop system for approximately one hour (making about 120 passes through the 0.2- μ m filter). The gas concentration in the test cell during the experiment is roughly 50 to 60% of that at saturation. As the cell is filled, hydrophobic polystyrene spheres with an average diameter of 0.245 μ m are added to the water. In this way, spheres that harbor potential cavitation nuclei are uniformly dispersed in the cell. For the measurements with Albunex, the test cell was filled with partially water only (no polystyrene spheres were added). Albunex was injected into the confocal regions of the transducers just prior to acquiring data rather than during the host fluid preparation because many of the bubbles rise to the surface within 15 min.

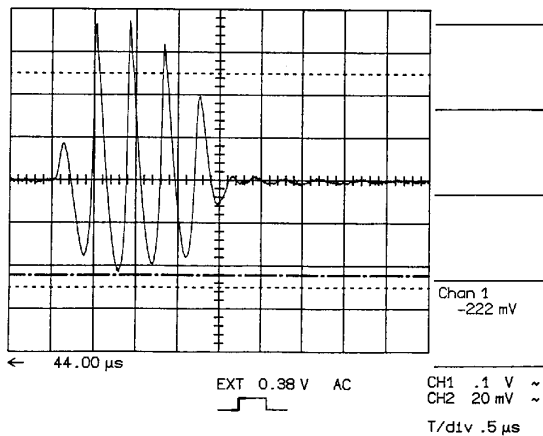
The role of the polystyrene spheres in the nucleation processes is not completely understood, but gas that has been adsorbed by the spheres most likely serves as the nucleation sites [17]. The fluid is allowed to sit about 30 min before the experiment commences so that any large bubbles that are entrained in the host fluid while filling the cavitation cell will float to the top of the cell. Because the polystyrene spheres are so small and are nearly neutrally buoyant, they remain



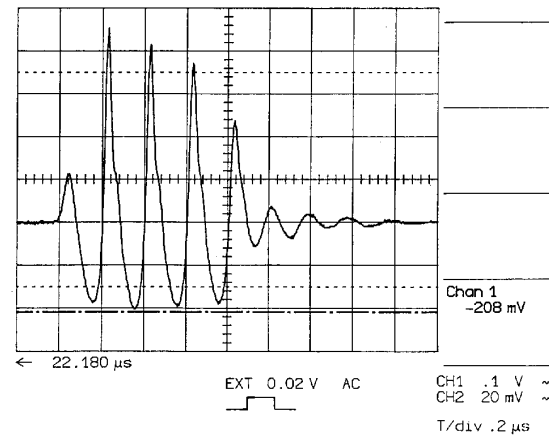
(a)



(a)



(b)



(b)

Fig. 4. Oscilloscope traces of the received PVDF signals for the 2.5-MHz HP transducer driven in (a) M-mode and (b) Doppler mode. The time base for both traces is $0.5 \mu\text{s}/\text{div}$.

Fig. 5. Oscilloscope traces of the received PVDF signals for the 5.0-MHz HP transducer driven in (a) M-mode and (b) Doppler mode. The time base for both traces is $0.2 \mu\text{s}/\text{div}$.

dispersed in the host fluid and do not settle to the bottom of the test cell. The concentration of polystyrene spheres employed for these experiments is 250×10^6 spheres/ml. During the experiment, acoustic streaming from the ACD, a second order effect, causes the polystyrene spheres to convect through the confocal region at a velocity of ~ 8 cm/s. Thus a constant stream of potential nuclei moves through the confocal region during the experiments.

IV. RESULTS AND DISCUSSION

Shown in Fig. 6 is a typical data record of backscattered signals from the ACD with $0.245\text{-}\mu\text{m}$ polystyrene spheres added to the water. The HP 2.5-MHz transducer was driven in M-mode by the HP ultrasound system with a peak negative pressure output of 1.3 MPa. The top trace is a condensed time base representation of all 20 bins of data. (One bin is one complete scope trace, or $50 \mu\text{s}$ of backscattering data.) It is possible to "leaf" through all of the bins and view them separately on an expanded time base scale. The

bottom two traces are expanded versions of the 5th and 18th bins, respectively. We attribute the $\sim 1 \mu\text{s}$ -duration signal to cavitation and the $\sim 10 \mu\text{s}$ -duration signal to a aggregate of polystyrene spheres. Note that the cavitation event lasts only as long as the M-mode pulse ($< 1 \mu\text{s}$). At this peak negative pressure level, 1.3 MPa, at least 1 bin out of 20 bins of data contained cavitation events for every data record. However, when the peak negative pressure amplitude was decreased to a value below 1.1 MPa, cavitation was not detected in over 20 data records. Thus a rough pressure "threshold" was established to be 1.1 MPa for 2.5 MHz. Backscattered signal amplitudes ranged from 120–400 mV for polystyrene sphere aggregates and from 160–800 mV for cavitation events.

Cavitation was also detected with the 2.5-MHz transmitter driven in Doppler mode, at an approximate threshold of 1.1 MPa. Fig. 7 contains traces of the received echoes from the cavitation produced. The top trace is the composite data record and the bottom trace is an expanded version of the 13th bin. The time base for the expanded signal is $2 \mu\text{s}/\text{div}$. Note that the

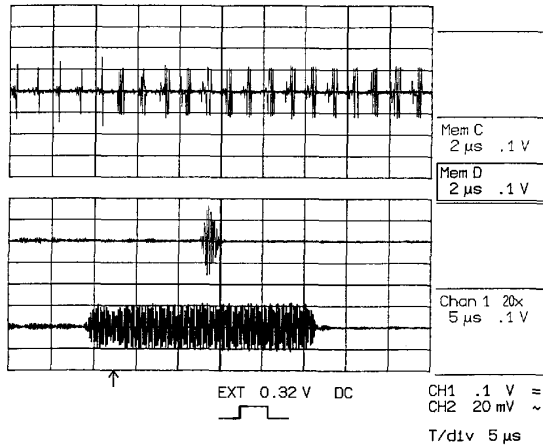


Fig. 6. Typical data record of backscattered signals from the ACD with 0.245 μm polystyrene spheres in water (HP scanner settings: 2.5-MHz transducer, M-mode, with output of 1.3 MPa peak negative pressure). The top trace is a condensed time base representation of one data record, or 20 scattering events. The middle trace is the expanded 5th bin, which we attribute to scattering from transient cavitation. The bottom trace is the 18th bin, which we attribute to scattering from polystyrene spheres. The time base of the two lower traces is 2 $\mu\text{s}/\text{div}$.

pulse duration of the 2.5-MHz Doppler waveform is 2 μs and that the duration of the cavitation event is also approximately 2 μs . Such an asymmetric envelope was a feature of all the scattered signals from cavitation events produced in the Doppler mode. Church has calculated the response of micron-sized, free bubbles in water to an acoustic pressure pulse similar to the Doppler waveform utilized by the HP ultrasound system [18]. The duration of the calculated bubble response is approximately the same as the duration of the Doppler pulse, although the bubble grows, collapses, and rebounds several times within the acoustic tone burst. The received, scattered signals from the cavitation events do not exhibit such growth and collapse cycles, perhaps because of 1) the detector's limited sampling rate of the bubble motion, or 2) the cavitation events that are detected involve multiple bubbles (e.g., the bubble shatters upon the first collapse and nucleates multiple cavitation events in the following cycles), or 3) the event was not nucleated from a simple, free bubble. Because of these possibilities, the direct application of Church's free-bubble calculations to our polystyrene sphere-nucleated cavitation should be executed with discretion.

Cavitation was not detected from the 5.0-MHz HP transducer in either M-mode or Doppler mode with polystyrene spheres added to the host fluid as potential nuclei. Fig. 8 contains a typical data record indicating the presence of an aggregate of polystyrene spheres in the focal region of the ACD. The bottom trace is a composite data record of 20 scattering events and the top trace is an expanded version of the 9th bin. The bins in this data record were recorded contiguously in time. Such a data record can be attributed to polystyrene spheres that were convected through the focal region of the ACD due to acoustic streaming.

When only Alburnex was added water, rather than polystyrene spheres, cavitation was not detected in either M-mode or

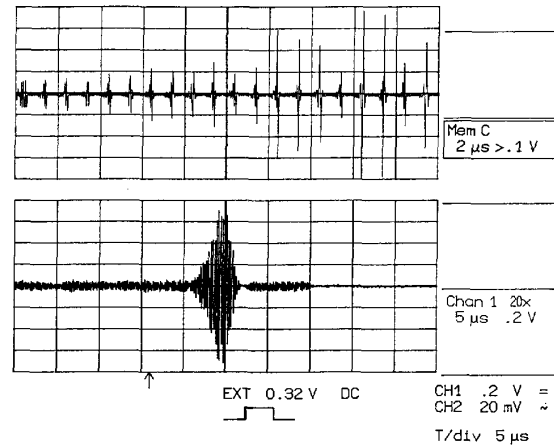


Fig. 7. Backscattered signals from cavitation produced with 2.5 MHz HP transducer driven in Doppler mode. The top trace is a composite data record of 20 scattering events and the bottom trace is an expanded version of the 13th bin. The time base for the expanded signal is 2 $\mu\text{s}/\text{div}$.

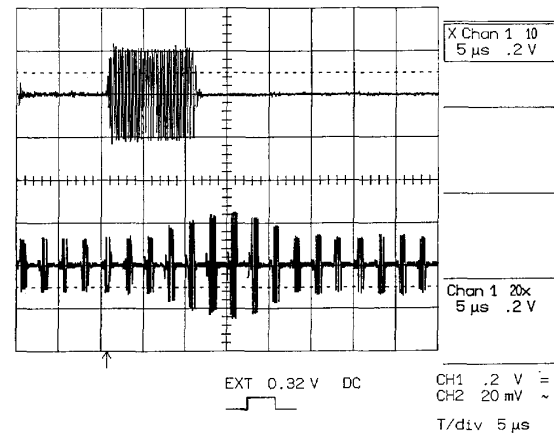


Fig. 8. Backscattered signals from an aggregate polystyrene spheres connecting through the focal region of the ACD. The bottom trace is a composite data record of 20 scattering events. The top trace is an expanded version of the 9th bin and its time base is 5 $\mu\text{s}/\text{div}$.

Doppler mode, even at the maximum drive amplitudes (peak rarefactional pressures of 1.3 MPa at 2.5 MHz and 1.2 MPa at 5.0 MHz). All of the received echoes had a pulse duration equal to that of the ACD pulse duration, 10 μs . Backscattered signal amplitudes ranged from 200–400 mV for the Alburnex (recall that cavitation events ranged from 160–800 mV in amplitude). It is unclear whether 1) the presence of 1–10 μm bubbles caused scattering events so large that cavitation events were obfuscated or 2) that the Alburnex did not nucleate bubble growth and collapse at 2.5 and 5.0 MHz. Previously published results obtained at lower frequencies and longer pulse lengths indicate that the threshold for cavitation of Alburnex in water ranges from 0.3 to 0.5 MPa peak negative pressure [5], [19].

Madanshetty *et al.* have measured cavitation thresholds in a range of 0.65 to 0.95 MPa at 0.75 MHz with the ACD under similar host fluid and nuclei conditions [20]. The lower

threshold, 0.65 MPa, was measured at a higher gas concentration (84% of saturation as opposed to 50–60%). The higher threshold, 0.95 MPa, was measured with a slightly smaller polystyrene sphere concentration (191×10^6 particles/ml rather than 250×10^6 particles/ml). The transducer that was used to produce the large amplitude pressure waves necessary to cause cavitation was a 1.5-inch diameter spherical segment of PZT8, a lead zirconate titanate ceramic. This transducer was driven with a 10 μ s tone burst, which allowed for about seven cycles at 0.75 MHz. Recall that an approximate threshold of 1.1 MPa was measured at 2.5 MHz for both the M-mode and Doppler pulses [21]. These results can be interpreted in the context of Holland and Apfel's approximate theory for the prediction of the onset of cavitation from single cycle ultrasound pulses. This theory predicts approximately a 0.1 MPa difference in the thresholds expected for 0.75-MHz and 2.5-MHz insonifying frequencies. It is not clear what effect the pulse duration, or the number of cycles in the pulse, has on the threshold for 1 to 10-cycle insonifying pulses.

A mechanical energy index has also been derived in order to describe the relationship between threshold pressure and frequency in a water- or blood-like host fluid with all sizes of nuclei present ($MI = P/f^{1/2}$) [22]. Below a value of $MI = 0.7$, it was argued that the physical conditions do not exist to promote bubble growth even in the presence of a broad nuclei distribution. The calculated values of the mechanical energy index for the approximate threshold measured at 2.5 MHz, 1.1 MPa, is 0.7. At the highest power setting, the 5.0-MHz HP transducer can produce a peak negative pressure of 1.2 MPa. The corresponding mechanical energy index value is 0.5. Thus the data presented in this paper give approximate thresholds that are consistent with an index value of 0.7. Also, for an index value below 0.7, cavitation is not detected.

V. CONCLUSION

The results presented in this paper suggest that the conditions for transient cavitation inception are satisfied in water seeded with polystyrene spheres and insonified with short-pulse diagnostic ultrasound at 2.5 MHz. However, these conditions are not met at 5.0 MHz for peak negative pressures less than 1.2 MPa. The extent of the cavitation events observed at 2.5 MHz was limited, so that a post-processing technique involving visualization of individual scattering events had to be utilized in order to discriminate polystyrene spheres from cavitation. Qualitative agreement exists between the bubble response theory of Church and the observed duration of these cavitation events. Cavitation was not detected in water seeded with Alunex at either frequency in M-mode or Doppler mode. *In vitro* measurements of cavitation thresholds in various biological fluids, such as blood and amniotic fluid, with the sensitive ACD system may provide information on the existence or absence of nuclei in these fluids *in vitro*.

Although cavitation, which was produced *in vitro* by an HP clinical diagnostic ultrasound scanner at 2.5 MHz, has been detected, our knowledge must be extended to the *in vivo* environment to ascertain direct clinical significance. The 30-MHz cavitation detection technique utilized in this work is

hampered by noisy environments, or "clutter" (scattering off nearby interfaces) and by attenuating media. Thus, only limited success may be realized for the detection of transient cavitation in the body with the 30-MHz ACD.

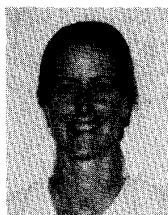
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REFERENCES

- [1] R. E. Apfel, "Acoustic cavitation: A possible consequence of biomedical uses of ultrasound," *Brit. J. Cancer*, vol. 45, suppl. 5, pp. 140–146, 1982.
- [2] H. G. Flynn, "Generation of transient cavities in liquids by microsecond pulses of ultrasound," *J. Acoust. Soc. Am.*, vol. 72, pp. 1926–1932, 1982.
- [3] A. A. Atchley, L. A. Frizzell, R. E. Apfel, C. K. Holland, S. Madanshetty, R. A. Roy, "Thresholds for cavitation produced in water by pulsed ultrasound," *Ultrason.*, vol. 26, pp. 280–285, 1988.
- [4] R. A. Roy, A. A. Atchley, L. A. Crum, J. B. Fowlkes, and J. J. Reidy, "A precise technique for the measurement of acoustic cavitation thresholds and some preliminary results," *J. Acoust. Soc. Am.*, vol. 78, pp. 1799–1805, 1985.
- [5] C. K. Holland and R. E. Apfel, "Thresholds for transient cavitation produced by pulsed ultrasound in a controlled nuclei environment," *J. Acoust. Soc. Am.*, vol. 88, pp. 2059–2069, 1990.
- [6] R. A. Roy, S. Madanshetty, and R. E. Apfel, "An acoustic backscattering technique for the detection of transient cavitation produced by microsecond pulses of ultrasound," *J. Acoust. Soc. Am.*, vol. 87, pp. 2451–2455, 1990.
- [7] S. C. Child, C. L. Hartman, L. A. Schery, E. L. Carstensen, "Lung damage from exposure to pulsed ultrasound," *Ultrasound Med. Biol.*, vol. 16, pp. 817–825, 1990.
- [8] M. S. Ziskin and D. B. Petitti, "Epidemiology of human exposure to ultrasound: A critical review," *Ultrasound Med. Biol.*, vol. 14, pp. 91–96, 1988.
- [9] P. L. Carson, P. R. Fischella, and T. V. Oughton, "Ultrasonic power and intensities produced by diagnostic ultrasound equipment," *Ultrasound Med. Biol.*, vol. 3, pp. 341–350, 1978.
- [10] F. A. Duck, H. C. Starritt, J. D. Aindow, M. A. Perkins, and A. J. Hawkins, "The output of pulse-echo ultrasound equipment: a survey of powers, pressures, and intensities," *Brit. J. Radiol.*, vol. 58, pp. 989–1001, 1985.
- [11] F. A. Duck, H. C. Starritt, and S. P. Anderson, "A survey of the acoustic output of ultrasonic doppler equipment," *Clin. Phys. Physiol. Meas.*, vol. 8, pp. 39–49, 1987.
- [12] R. E. Apfel, "Acoustic cavitation," in *Methods of Experimental Physics*, P. Edmonds, Ed. New York: Academic, 1981, vol. 19, ch. 7, p. 385.
- [13] H. G. Flynn, "Cavitation dynamics—I. A mathematical formulation," *J. Acoust. Soc. Am.*, vol. 57, pp. 1379–1396, 1975.
- [14] Q. Xu and R. E. Apfel, "The detection of microbubble radius versus time and some comparisons with theoretical results," *J. Acoust. Soc. Am.*, vol. 87, p. S141, 1990.
- [15] H. C. Starritt, M. A. Perkins, F. A. Duck, and B. F. Humphrey, "Evidence for ultrasonic finite-amplitude distortion in muscle using medical equipment," *J. Acoust. Soc. Am.*, vol. 77, pp. 302–306, 1985.
- [16] T. A. Siddiqi, R. A. Meyer, J. Sullivan, M. Miodovnik, C. Daft, and W. D. O'Brien, Jr., "In vivo ultrasound exosimetry: The ovarian examination," *Med. Phys.*, vol. 17, p. 548, 1990.
- [17] C. K. Holland, "Thresholds of transient cavitation produced by pulsed ultrasound in a controlled nuclei environment," Ph.D. dissertation, Yale Univ., New Haven, CT, 1989, (available from University Microfilms International, 300 N. Zeeb Road, Ann Arbor, MI 48106).

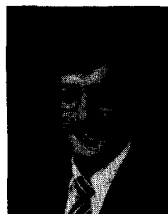
- [18] C. C. Church, "A comparison between 'real' and 'ideal' gas in theoretical cavitation dynamics," *J. Acoust. Soc. Am.*, vol. 89, no. 4, pt. 2, pp. 1862–1863, 1991.
- [19] R. A. Roy, C. C. Church, and A. Calabrese, "Cavitation produced by short pulses of ultrasound," in *Frontiers of Nonlinear Acoustics*, 12th ISNA, M. F. Hamilton and D. T. Blackstock, Eds. New York: Elsevier, 1990.
- [20] S. I. Madanshetty, R. A. Roy, R. E. Apfel, "Acoustic microcavitation: Its active and passive acoustic detection," *J. Acoust. Soc. Am.*, vol. 90, pp. 1515–1526, 1991.
- [21] C. K. Holland and R. E. Apfel, "An improved theory for the prediction of microcavitation due to pulsed ultrasound," *IEEE Trans. Ultrason., Ferroelec., Freq. Contr.*, vol. 36, pp. 204–208, 1989.
- [22] R. E. Apfel and C. K. Holland, "Gauging the likelihood of cavitation from short pulse, low duty cycle diagnostic ultrasound," *Ultrasound in Med. and Biol.*, vol. 17, pp. 179–185, 1991.



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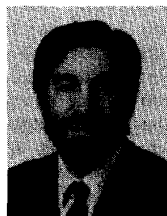
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